

C04 Assessment Tool-1(Data Interpretation)

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Course Code: ITA0502-Computer Vision

1.Real-Time Detection System – Comparison and Selection

A surveillance system requires both high detection accuracy and low processing latency. The three techniques can be compared as follows:

Method	Accuracy	Speed	Suitability
Viola-Jones	78%	Fast	Good for simple, low-cost systems
YOLO	92%	Moderate	Best for real-time surveillance
Deep Learning	95%	Slow	Best when accuracy is more important than speed

1. Viola-Jones

Viola-Jones is a traditional object/face detection technique. It is very fast and requires low computational power, making it suitable for low-resource devices. However, its 78% accuracy is comparatively low and it may perform poorly under different lighting, poses, occlusion, and complex backgrounds.

2. YOLO

YOLO is a modern object-detection method that provides 92% accuracy with moderate speed. It can detect multiple objects in real time and can be optimized using GPUs, NPUs, quantization, or lightweight models. It provides a good balance between accuracy and latency.

3. Deep Learning

Deep Learning provides the highest accuracy of 95%, but it is the slowest method in the given comparison. It requires more computational resources and may introduce unacceptable latency in a real-time surveillance system.

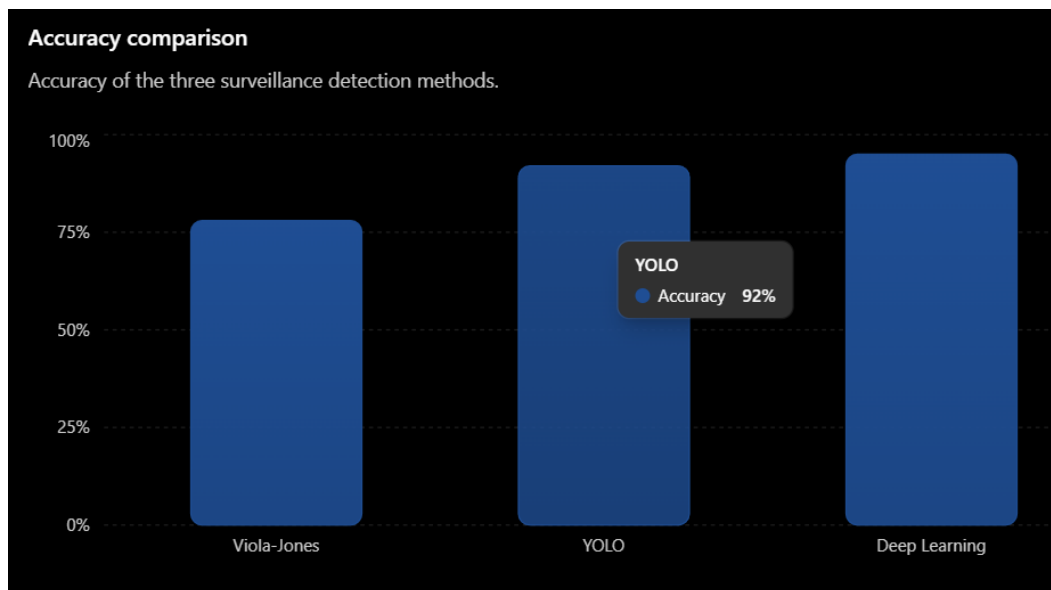


Fig 1:Accuracy Comparison

Selection: YOLO

YOLO should be selected because it provides the best trade-off between speed and accuracy.

- Accuracy: 92%, much better than Viola-Jones.
- Speed: Moderate, making it suitable for real-time applications.
- Computational requirements: Higher than Viola-Jones but manageable with modern hardware.
- It can be deployed on edge devices using hardware acceleration.
- The 3% accuracy advantage of Deep Learning may not justify its additional latency.

Deployment Conditions

- Low-power device: Viola-Jones can be used if accuracy requirements are low.
- Edge device with GPU/NPU: YOLO is the preferred choice.
- High-performance server with no strict latency: Deep Learning may be considered.
- Security-critical real-time surveillance: YOLO is the most suitable option.

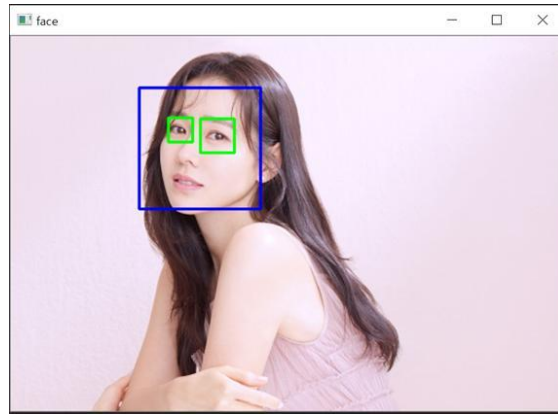
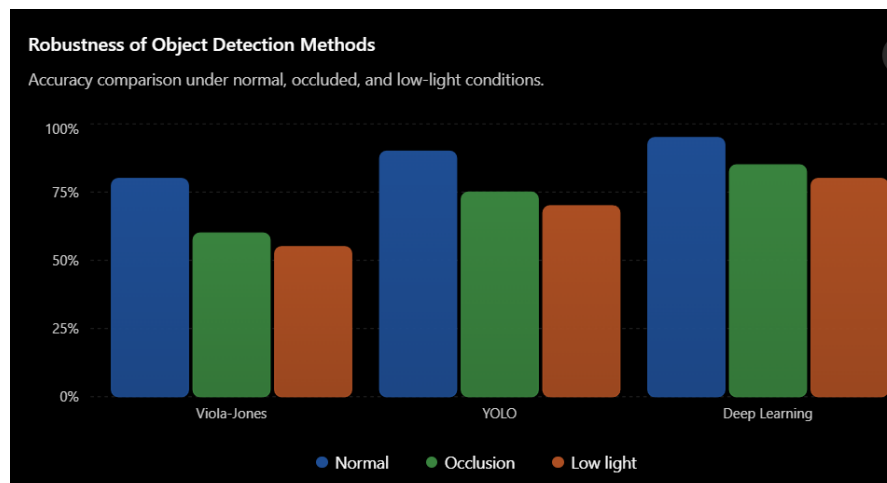


Fig 1:YOLO Object Detection

YOLO is the recommended method for a surveillance system with strict latency constraints. It achieves 92% accuracy with moderate processing speed, providing a better balance than Viola-Jones and Deep Learning. A lightweight YOLO model with GPU/NPU acceleration can further reduce latency while maintaining good detection accuracy.

2. Robust Object Detection

The robustness of an object-detection method can be evaluated by checking its performance under normal conditions, occlusion, and low-light conditions.



1. Viola-Jones

Viola-Jones performs reasonably well under normal conditions with 80% accuracy. However, its performance drops significantly to 60% under occlusion and 55% in low-light conditions.

This shows that it is less robust when objects are partially hidden or lighting conditions are poor. Its main advantage is fast processing and low computational requirements.

2. YOLO

YOLO achieves 90% accuracy under normal conditions, 75% under occlusion, and 70% under low light.

It is more robust than Viola-Jones because it can handle complex scenes and partial object visibility better. It also provides a good balance between accuracy, speed, and computational requirements.

3. Deep Learning

Deep Learning provides the best results in all three conditions:

- Normal: 95%
- Occlusion: 85%
- Low light: 80%

It loses only 10 percentage points from normal to low-light conditions, showing strong robustness. However, this comes with higher computational requirements and slower processing.

Comparison

Method	Normal	Occlusion	Low Light	Overall Robustness
Viola-Jones	80%	60%	55%	Low
YOLO	90%	75%	70%	Good
Deep Learning	95%	85%	80%	Best

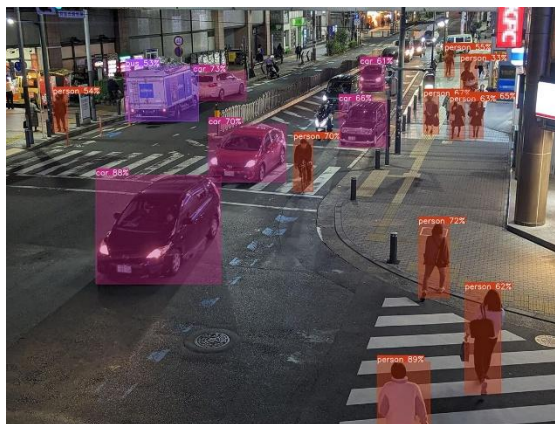
If we take the average performance:

- Viola-Jones: $(80 + 60 + 55) / 3 = 65\%$
- YOLO: $(90 + 75 + 70) / 3 = 78.3\%$
- Deep Learning: $(95 + 85 + 80) / 3 = 86.7\%$

Best Choice: Deep Learning

Deep Learning is the best choice for robust object detection because it achieves the highest accuracy under all conditions:

- Normal: 95%
- Occlusion: 85%
- Low Light: 80%
- Average: 86.7%



Deep Learning-Based Robust Object Detection Under Occlusion and Low-Light conditions

It is more reliable when objects are partially blocked or when lighting is poor.

Trade-off: It requires more computational power and is slower than YOLO and Viola-Jones.

Final decision: If the main priority is robustness and accuracy, choose Deep Learning. If real-time speed is more important, choose YOLO.

3. Optical Flow Analysis

Optical flow is used to estimate the motion of objects between consecutive video frames. The two methods have different advantages and limitations.

Factor	Method A	Method B
Motion detection	Misses small motions	Detects fine motions
Stability	High	Low

Noise sensitivity	Low	High
Accuracy for large motion	Good	Good
Accuracy for small motion	Low	High
Motion vectors	Stable and smooth	Noisy
Dynamic scene suitability	Good	Moderate
Overall	Stable	Sensitive but detailed

Method A – Stable Motion Vectors

Method A produces stable and consistent motion vectors. This makes it less sensitive to noise and suitable for scenes with large or continuous movements.

However, it may fail to detect small or subtle movements, reducing its accuracy when objects move slowly or when fine motion is important.

Advantages:

- Stable motion estimation
- Low noise sensitivity
- Reliable for large movements
- Suitable for real-time applications

Disadvantage:

- May miss small movements.

Method B – Fine Motion Detection

Method B is capable of detecting small and fine movements, making it more sensitive and potentially more accurate for detailed motion tracking.

However, it produces noisy motion vectors, which can lead to false motion detection and unstable tracking.

Advantages:

- Detects small movements
- High sensitivity
- Better for detailed motion analysis

Disadvantages:

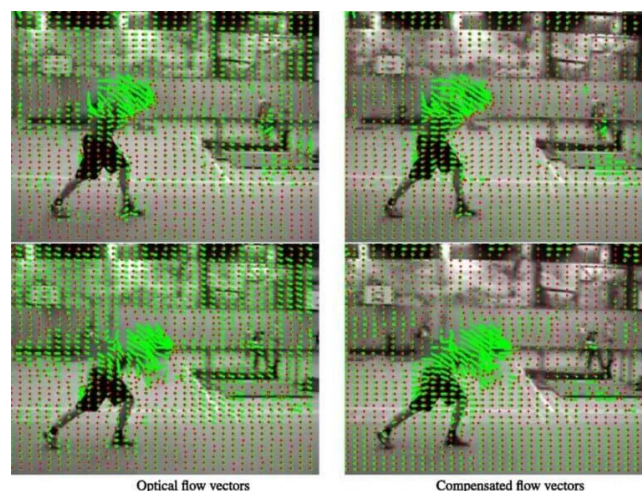
- More sensitive to noise
- Motion vectors may be unstable
- May require filtering or smoothing.

Best Choice: Method A

For general motion tracking in dynamic scenes, Method A is the better choice because its stable motion vectors provide more reliable tracking and are less affected by noise.

For example, in traffic surveillance, video monitoring, and object tracking, stability is often more important than detecting extremely small movements.

However, Method B should be selected when the application specifically requires detection of fine or subtle movements, such as gesture recognition, medical motion analysis, or detailed activity monitoring.



Comparison of Stable and Noisy Optical Flow Motion Vectors in Dynamic Scenes

Method A is recommended for dynamic-scene motion tracking because it provides stable, reliable, and noise-resistant motion vectors. Although it may miss small motions, its stability makes it more suitable for practical real-time tracking. Method B is preferable only when detecting small movements is the primary requirement.

4. Model Generalization and Overfitting

Model generalization refers to how well a machine learning model performs on unseen data. A good model should have high testing accuracy and only a small difference between training and testing performance.

Model	Training Accuracy	Testing Accuracy	Train-Test Gap	Overfitting Risk
A	98%	70%	28%	Very High
B	90%	88%	2%	Low

Model A

Model A achieves 98% training accuracy, which initially appears excellent. However, its testing accuracy drops to only 70%.

The difference is:

$98\% - 70\% = 28\%$

This large gap indicates that Model A has overfitted the training data. It has learned the training examples too specifically instead of learning general patterns that can be applied to new data.

Problems with Model A:

- Very high overfitting risk
- Poor performance on unseen data
- Weak generalization ability
- Unreliable for real-world deployment

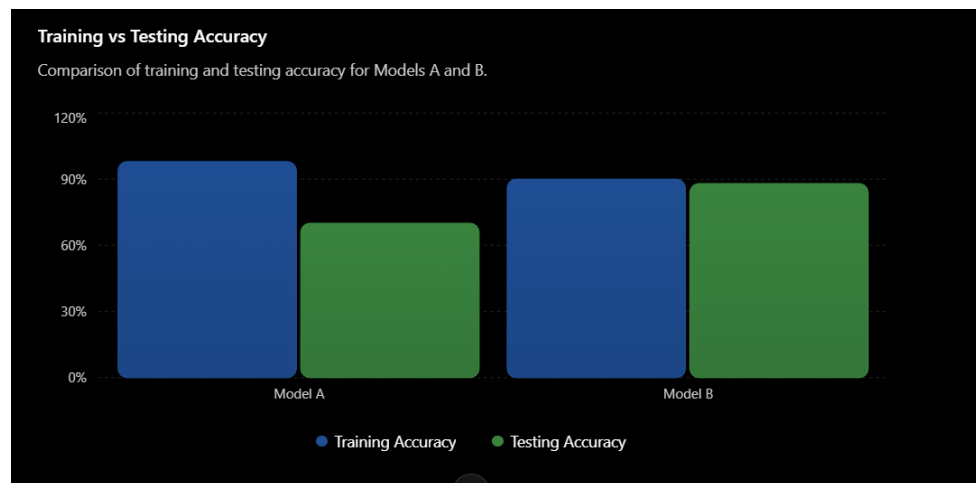
Although Model A performs extremely well on training data, its low testing accuracy makes it unsuitable for practical applications.

Model B

Model B has 90% training accuracy and 88% testing accuracy.

The difference is: $90\% - 88\% = 2\%$

The small gap indicates that Model B performs consistently on both training and unseen testing data.



Advantages of Model B:

- Low overfitting risk
- Strong generalization ability
- Consistent performance
- More reliable on unseen data
- Better suited for real-world deployment

The slightly lower training accuracy is not a disadvantage because the model maintains high accuracy on testing data.

Generalization Comparison

Model A:

Training → 98%

Testing → 70%

Large gap → Poor generalization

Model B:

Training → 90%

Testing → 88%

Small gap → Good generalization

Therefore, Model B has much better generalization ability.

Best Choice: Model B

Model B should be selected for real-world deployment.

The main reason is its 88% testing accuracy and only 2% training-testing gap. This indicates that Model B has learned general patterns rather than memorizing the training data.

Model A has a higher training accuracy of 98%, but its testing accuracy is only 70%. In real-world applications, performance on unseen data is more important than performance on training data.

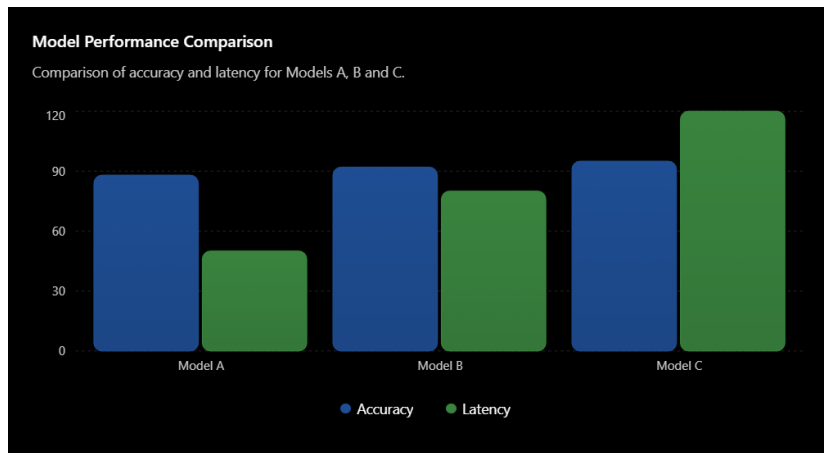
Model B is more reliable and should be deployed. It demonstrates strong generalization, low overfitting, and consistent performance between training and testing data. Model A is highly overfitted because of its 28% accuracy gap, making it less reliable for real-world use.

Final answer: Model B → Better generalization + Low overfitting + Higher reliability.

5. System Performance Under Constraints

A system should be selected by considering **accuracy, latency, and cost together**. A model with the highest accuracy is not always the best choice if it has high latency and cost.

Model	Accuracy	Latency	Cost	Overall Evaluation
A	88%	50 ms	Low	Fast and economical
B	92%	80 ms	Moderate	Best balance
C	95%	120 ms	High	Highest accuracy but expensive and slow



Model A

Model A has 88% accuracy, the lowest latency of 50 ms, and low cost.

Advantages:

- Fast response
- Low computational cost
- Suitable for real-time systems

Disadvantage:

- Lower accuracy compared with B and C.

Model A is suitable when speed and low cost are the highest priorities.

Model B

Model B provides 92% accuracy, 80 ms latency, and moderate cost.

It improves accuracy by 4 percentage points over A, while the latency increase is only 30 ms.

It also provides a significant cost advantage compared with Model C.

Therefore, Model B provides a strong compromise between all three factors.

Model C

Model C achieves the highest accuracy of 95%, but it has the highest latency (120 ms) and high cost.

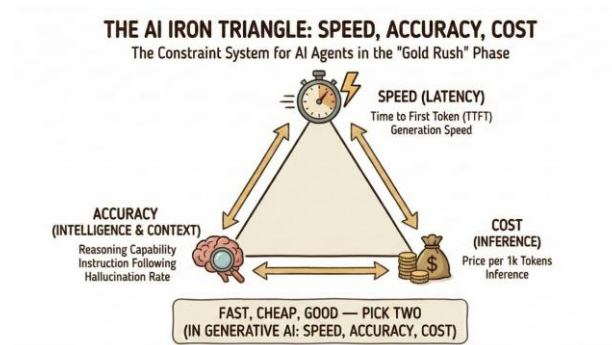
It is suitable when maximum accuracy is more important than response time and budget.

However, for a system with practical resource constraints, the additional 3% accuracy over Model B may not justify the increased latency and cost.

Best Choice: Model B

Model B is the best overall choice because it provides:

- High accuracy: 92%
- Acceptable latency: 80 ms
- Moderate cost
- Good balance between performance and resource usage



Accuracy, Latency, and Cost Trade-off in Real-Time AI Systems

Model B should be selected when the objective is to achieve the best balance between accuracy, latency, and cost. Model A is preferable for highly cost-sensitive and latency-critical systems, while Model C is suitable when maximum accuracy is the primary requirement.

Final decision: Model B → Best overall balance.

